The Lifelong Loop: Error-Gated Consolidation and the Cost of Nights Without a Log

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Abstract

Can an agent whose long-term memory lives in the weights of a frozen language model keep learning, day after day? We run a complete day/night memory loop end-to-end for up to 30 simulated days (720 facts): each day streams facts into a disposable LoRA "day" adapter with error-gated writes; each night consolidates into a persistent "core" adapter and resets the day store; after every night, per-fact recall and 2-AFC recognition are probed over the *entire* history. Five findings, all with a capability firewall (adapter-off equals base) intact in every run. (1) At *identical* nightly budget, **error-gated consolidation** — train the core only on what it currently fails — retains 2.2× more than recency-only consolidation and flattens the retention-by-age curve. (2) The class that untargeted consolidation loses is precisely the *latent* one: facts recognized-but-not-recalled have 0% next-day survival under recency-only consolidation (0/588) and 46% under error-gating — above even the recalled class, because the miss list targets them. (3) **Nights without a log do not merely fail; they poison**: pure self-recitation collapses structurally (0/144 — the core cannot recite what only the day adapter learned), and mixing recitation with ground truth is *worse than doing nothing about the past* (4.3/144, below the recency baseline's 25.0), because wrongly-recited old facts overwrite earlier consolidation — error compounding, a within-system analogue of model collapse. (4) Over a 30-day horizon at fixed budget, **neither anticipated exhaustion occurs**: full replay holds 99% of 720 facts (the r32 core's streaming capacity exceeds the stream our generator can produce), and the error-gated arm's retained *fraction* rises with horizon (38% → 44% → 46%), settling into a throttled steady state — a stable old-day plateau bought by squeezing intake of the newest days. (5) The served agent is **not aphasic** — GSM8K is intact with adapters mounted in both serving states — but free-text fluency shows a graded, super-additive "accent" that concentrates in the hot-day serving state, the same state responsible for a 3–10× *read-time masking* of old facts. Together these results yield a concrete design rule for log-backed, self-testing in-weight memory: consolidate nightly on failures, serve from the core, mount the day adapter narrowly — and never let the system rehearse its own unverified recall.

1 Introduction

Deployed LLM agents accumulate experience, but their weights do not: memory is delegated to context windows, retrieval stores, and files, while the model itself stays frozen. A growing line of work proposes to close that gap with a sleep-like loop — act during the day, consolidate into parameters at night (Sleep-cycle agents, 2026; Sleep-consolidated memory, 2026; Behrouz et al., 2024). The proposals are architectural; what the field lacks is *measurement*: when such a loop actually runs for many days, what survives, what is lost, what does each night's policy cost, and does the host model remain usable?

In companion work (Lin, 2026) we audited the *day* scale: single-pass fact streams written into a LoRA adapter on a frozen base, instrumented per fact. Three results from that audit drive the present design. First, online in-weight memory accumulates by *re-instatement*, not persistence — capacity is determined by what gets re-trained after it slips, so the schedule of re-instatement is the central design variable. Second, forgetting under protected writes is largely *suppression* rather than erasure: a cheap two-alternative recognition probe shows most "forgotten" facts remain latent, which makes targeted re-instatement cheap. Third, spending a fixed replay budget on *self-test failures* converts those latent facts back to expressed ones at no extra cost.

This paper closes the loop. We run a complete day/night system — disposable day adapter, persistent core, ground-truth log, nightly consolidation, daily reset — for 6, 12, and 30 simulated days, and we treat the night policy as the experimental variable under a fixed gradient budget. The questions are the ones a lifelong-learning deployment would actually face: Should the night replay everything (cost grows without bound), only the most recent day (cheap, but what of the past?), only what the core currently fails (cheap and targeted, but the failure list grows), or the core's own recitations (no log dependency at all)? What breaks first as the horizon grows — the budget, or the core's capacity? And is the agent still a usable language model while all of this is mounted?

Contributions.

1. **An end-to-end loop instrument** (§3): per-fact recall and recognition over the entire history after every night, retention-by-age curves, category-conditioned survival (recalled / latent / gone), dual-state capability probes, and a capability firewall checked in every run — to our knowledge the first per-fact audit of a multi-day consolidate-into-weights loop.

2. **Budget-matched night policies** (§4.1): error-gated consolidation retains 2.2× recency-only consolidation at identical budget (55.3 vs. 25.0 of 144) and flattens the age curve; the mechanism is located in the *latent* class, whose next-day survival is 0% without targeting and 46% with it.

3. **The log is load-bearing — twice** (§4.2): in a day/core split the core cannot recite what only the day adapter learned (pure self-recitation: 0/144), and it cannot *safely* recite what it partially learned: a pre-registered hybrid (recite old + log for today) landed *below* the recency baseline (4.3 vs. 25.0) because wrong recitations actively overwrite earlier consolidation. Active wrong-value rehearsal is worse than passive decay.

4. **Horizon behavior at fixed budget** (§4.3): across 6 → 12 → 30 days (144 → 720 facts), the error-gated arm's retained fraction *rises* (38/44/46%) and settles into a throttled steady state (old-day plateau, squeezed intake); full replay holds 99% at 720 facts, so the r32 core's streaming capacity exceeds what our fact generator can produce. Neither the pre-registered budget collapse nor capacity exhaustion occurs; both knees lie beyond this toy's reach.

5. **Dual-state capability and read-time masking** (§4.4): mounted adapters leave GSM8K intact in both serving states (no task-level aphasia), but free-text NLL rises super-additively

under the hot day mount (∼6× perplexity), and the same state masks old-fact recall 3–10× at read time. The serving prescription follows: consolidate nightly, serve from the core, mount the day adapter narrowly.

As in the companion paper, everything is reported as floors and directions — one substrate, synthetic facts, 1–3 seeds per condition on nondeterministic consumer hardware — with per-seed numbers throughout and raw per-fact timelines released.

2 Related work

Sleep-cycle agents. Architectures that batch daytime experience and consolidate into parameters at rest are now numerous — adaptive agents with sleep phases (Sleep-cycle agents, 2026), sleep-consolidated memory (Sleep-consolidated memory, 2026), and test-time memorization lines (Behrouz et al., 2024). These works build and assert; we instrument and measure a running loop per fact, per night, per age — and several of their implicit assumptions (that self-distillation suffices; that serving with all adapters mounted is free) fail measurably here. **Complementary learning systems.** The day/core split follows CLS (McClelland et al., 1995): a fast, disposable store and a slow, protected one, with consolidation between. Our results give the computational version two sharp caveats: consolidation must be driven by ground truth (not replayed dreams of it), and the fast store interferes with the slow one at *read* time, not just write time. **Replay and selection.** Prioritized and interference-aware replay (Schaul et al., 2015; Aljundi et al., 2019), surprise-gated storage (Hazard et al., 2025), and forgetting-curve schedules (Lu et al., 2026) motivate the error-gated night; as in the companion paper we claim no novelty for the selection rule itself — the contribution is the budget-matched measurement of what it buys (and what its self-test costs) inside a full loop. Pseudo-rehearsal (Robins, 1995) anticipated consolidation without stored data; our recite and hybrid arms are its LLM-scale test, and they fail in an instructive way: generation quality is now high enough that wrong recitations are *fluent*, so they train as effectively as truth — which is precisely the problem. The collapse dynamics echo training-on-own-outputs degradation (Shumailov et al., 2024) inside a single system. **Behavioral horizon scaling.** Long-horizon environment learning now has behavioral scaling laws (ByteDance Seed, 2026); the substrate-level question underneath — what happens to in-weight memory as the horizon grows — is the gap this paper measures.

3 The loop instrument

Frozen Qwen3.5-2B base (fp32, eager attention); persistent rank-32 CORE adapter + disposable rank-64 DAY adapter (both all-linear LoRA, α=2r; AdamW, lr 3×10^{-5}). Facts are collision-free *"The user's {attribute} is {pseudoword}."* statements (the day-scale instrument of Lin, 2026); 24 facts per day.

Day. Each fact: 8 value-masked write steps into DAY with summed-Fisher EWC (λ=300), plus 4 replay steps spent on the most recent self-test's failures (the day-scale winner). Serving state during the day: CORE+DAY.

Night. Consolidate into CORE, then delete and re-initialize DAY. The gradient budget is fixed at $72 = 24 \times 3$ steps per night for every arm except *full*:

- *full*: all past true statements, 3 epochs — budget grows with history; the reference ceiling, deliberately **not** budget-matched;

night arm	final recall /144 (seeds)	mean	final recognition /144	next-day survival R/L/G
full (ceiling)	134/142/133	136.3	144/144/143	93% / 80% / 67%
misslog	52/60/54	**55.3**	117–125	36% / **46%** / 30%
markov	26/25/24	25.0	89–107	14% / **0%** / 0%
hybrid	6/6/1	4.3	72–89 (≈chance)	—
recite	0/0/0	0.0	55–76 (≈chance 72)	—

Table 1: Six-day loop (144 facts), post-night core-only recall of the full history. R/L/G = next-day survival of facts that were recalled / latent (recognized-only) / gone at the previous probe, pooled over seeds and days. Firewall 0.70 → 0.70 in every run.

- *markov*: today's true statements only — the scalable recency baseline;
- *misslog*: probe CORE-alone recall over the whole history (the mechanism's self-test; a generation cost that grows linearly with history, not a gradient cost), then train only on current failures, from the log;
- *recite*: self-distill the core's own greedy completions of the whole history — no log;
- *hybrid*: today from the log + old facts self-recited, budget spread uniformly.

Probes. After every night (every third night at the 30-day horizon), CORE-only greedy recall and 2-AFC recognition margins over *all* facts so far; the distractor for each fact is another fact's value from the same stream, so a positive margin measures fact-specific binding. **Firewall**: GSM8K (10-item fixed subset) with all adapters disabled, before and after every run. **Dual-state capability** (§4.4): GSM8K and fixed-text NLL/token with CORE+DAY mounted (end of day) and CORE alone (after night). Seeds 1234/2025/777 (D6), 1234/2025 or 1234 (D30); consumer ROCm GPUs are nondeterministic at fixed seed, so all numbers are directions with per-seed values shown. Fact cohorts at equal seed are prefix-identical across horizons, so D6/D12/D30 comparisons are exact.

4 Results

4.1 Night policy at matched budget

Table 1: at identical nightly budget, error-gated consolidation retains 2.2× the recency baseline (55.3 vs. 25.0 of 144), and the age curves (Fig. 1a shows the 30-day version) differ in kind: markov keeps only a ∼2-day recency window; misslog holds a plateau across all past days, paid for by lower retention of the newest days.

The survival-by-category columns locate the mechanism. Under markov, facts that were *latent* — recognized but not recalled, the suppressed class that dominates in-weight forgetting (Lin, 2026) — have **zero** next-day survival (0/588): each further night of recency-only training erases precisely the class that was cheapest to save. Under misslog the same class survives at 46%, *above* the recalled class (36%), because the self-test's miss list preferentially targets it. Error-gated consolidation is, operationally, a latent-fact harvester.

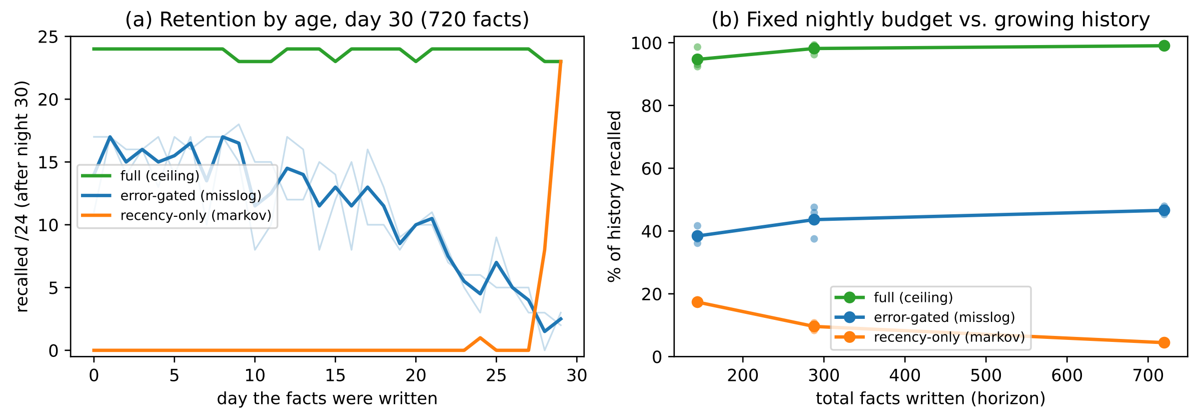

Figure 1: (a) Retention by write-day after night 30 (720 facts; thin lines = seeds, thick = mean). Full replay is flat at ceiling; error-gated consolidation holds an old-day plateau and pays with throttled intake of the newest days; recency-only keeps a two-day window. (b) Fraction of the whole history recalled vs. horizon at fixed nightly budget: the error-gated fraction *rises* (38%→44%→46%) while recency-only collapses (17%→10%→4%).

4.2 Nights without a log: recitation does not just fail, it poisons

Pure self-recitation collapses structurally (0/144, recognition at chance): in a day/core split, the core cannot recite content that only the day adapter learned, so there is nothing meaningful to distill and the night trains on noise. This is an architectural fact — single-core designs, where the core itself day-learns before reciting, degrade far more gracefully — and it is one half of why the log is load-bearing.

The *hybrid* arm was pre-registered (2026-07-08) to rescue this: today's facts from the log (solving the structural problem), old facts recited, same budget — our prediction placed it between markov and misslog. **The prediction was falsified downward: 4.3/144, below markov.** Per-cohort traces show why. Hybrid's today-cohort consolidates normally at its own night (7/9/4 of 24, comparable to markov) and is then *erased* over subsequent nights, reaching ∼0 by day 6, while markov's cohorts merely decay passively (19–22 of 24 survive one night). The difference: hybrid *actively rehearses wrong values.* Each night it recites old clozes mostly incorrectly — fluently, plausibly, in exactly the training format — and gradient descent cannot tell fluent errors from truth. The errors overwrite earlier correct consolidation and compound across nights: a within-system analogue of model collapse (Shumailov et al., 2024), and the second half of why the log is load-bearing. **If ground truth is unavailable for the past, doing nothing about the past (markov) beats self-distilling it (hybrid) by 6×.**

4.3 Horizon: fixed budget vs. growing history

We pre-registered (2026-07-08) three horizon predictions: (a) full replay stays ≥90% unless the r32 core saturates; (b) markov keeps a constant absolute window; (c) the error-gated arm's fraction falls below its 6-day value as the fixed budget races the growing miss list — with the shape of the decline as the finding. Outcomes at 12 days (288 facts) and 30 days (720 facts, Fig. 1): (a) *held* — 98% at 288, **99% at 720 (713/720, recognition 720/720, every day at 22–24/24)**: the core's streaming capacity exceeds what our collision-free generator can produce

(784 unique attributes), so the capacity knee is beyond this toy's reach; (b) *held exactly* — 32/720 (4.4%), the last ∼2 days; (c) **falsified in the favorable direction** — the error-gated fraction *rose* across horizons (38% → 44% → 45–48%). The by-day curves show the regime it settles into: an old-day plateau at 8–17/24 with intake of the newest ∼5 days squeezed to 2–6/24. At fixed budget the system does not race to a cliff; it **throttles intake to protect its stock** — a stable steady state in which final recognition is still 639–646/720 (∼89%), so the latent pool continues to dwarf expressed recall even after a month of nights. Locating the true knee requires harsher ratios (more facts per night-step) or a larger generator; both are future work, and we note plainly that our fact generator was exhausted before the system was.

4.4 Is the served agent still functional? Dual-state capability and read-time masking

All metrics above are cloze completions; a deployment needs the mounted agent to remain a usable language model. Re-running the 6-day error-gated arm with capability probes in both serving states (2 seeds) answers this. **Structured tasks are intact in both states**: GSM8K is 0.70–1.00 with the core alone (morning state) and 0.70–0.90 with the hot day adapter mounted (daytime state), vs. base 0.70 — 24/24 measurements at or above base. No task-level aphasia. **Free-text fluency shows a graded accent**: NLL/token on fixed neutral text rises from 2.93 (base) to 3.21–3.53 with the core alone — growing slowly with consolidated days — and to 4.4–5.0 under the hot day mount (∼6× perplexity), super-additively (each adapter alone is mild).

The hot day mount is also the state that damages *memory* readout. Comparing each night's pre-consolidation probe (CORE+DAY) with its post-consolidation probe (CORE only) exposes a systematic gap at every horizon in every arm: at the 30-day final night the error-gated run recalls 70/720 with the hot day adapter attached and 326/720 without it; even full replay shows 678 vs. 713. A freshly-written day adapter dominates the readout and emits its own day's values for old prompts: **concurrent adapter serving taxes old-fact recall** 3–10×. Both pathologies concentrate in one serving state, which yields the deployment prescription: *consolidate nightly; serve from the core; mount the day adapter narrowly (for recent-information queries) or attenuated — not permanently.*

5 Discussion

A design rule, stated once. The loop's results compose into a single sentence: a lifelong in-weight memory should be *log-backed* (recitation poisons), *self-testing* (the latent class is invisible to recency policies and free to harvest), *error-gated at both timescales* (day writes and night consolidation), and *served from its slow store* (the fast store taxes both fluency and old recall at read time). Every clause is a measured comparison, not a preference. **What the steady state means.** The throttled regime — protect the stock, squeeze the intake — is not a failure mode; it is what bounded consolidation *should* do, and it mirrors the classic stability–plasticity trade-off surfacing at the systems level. But it implies a design obligation: intake throttling must be made visible (the system's newest knowledge is its weakest), e.g. by routing recent-information queries to the day adapter or the log rather than the core. **What would falsify the rule.** A consolidation policy that beats misslog at matched budget without a log (contradicting §4.2), or a horizon at which the throttled state collapses rather than plateaus (contradicting §4.3), or a substrate where recitation is high-fidelity enough to avoid poisoning. All three are runnable with the released instrument.

6 Limitations

One substrate (Qwen3.5-2B) at the loop level; the companion paper's substrate-transfer results cover only the day scale. Synthetic, collision-free, non-conflicting facts; 24 facts per day is a toy day; revision ("the user moved") is untested in the loop. One nightly budget (72 steps) and one day/core rank pair; the throttled steady state's location presumably moves with both. 1–3 seeds per condition on nondeterministic hardware — directions, not calibrated points. The misslog self-test costs generation time linear in history; a deployment would subsample it, and whether sampled self-tests preserve the steady state is untested. The hybrid arm spreads its budget uniformly (today's per-fact share shrinks with history); a weighted variant could soften — though, per the cohort traces, not eliminate — the poisoning. Capability probes are GSM8K (10 items) plus a fixed-text NLL meter; free-form dialogue quality was not scored. Recognition margins use same-stream distractors; margins near zero are not calibrated confidence.

7 Conclusion

Run end-to-end for a month of simulated days, a frozen-base day/core memory system neither collapsed nor ran out of room — but only under a specific discipline: ground truth in the loop every night, consolidation targeted at what the system itself fails, and serving from the slow store. Relax any clause and the failure is measurable and mechanistically legible: skip the log and the system poisons itself with its own fluent errors; skip the targeting and the latent majority of memory dies silently; serve from the fast store and both fluency and old recall pay multiples. The loop's economics settle into a throttled steady state — stock protected, intake squeezed — whose knee lies beyond what our toy could reach. The per-fact, per-night instrument is released so that the knee, the substrates, and the policies we did not test can be measured rather than asserted.

AI disclosure

As in the companion paper: Claude Fable 5 and Claude Opus (Anthropic) implemented and ran the experiments, performed related-work searches, and drafted the manuscript under the author's direction; research questions, designs, pre-registrations, and final claims were set and approved by the author; all reported numbers re-derive from the released raw timelines.

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A Reproduction

Driver: `loop.py` (night arm, days, budget, probe stride, dual-state capability probes, core checkpointing); raw per-night, per-fact timelines: `results/loop_*.json`; `analyze.py` re-derives every table and `make_figure.py` the figure. The day-scale instrument, its results, and the companion paper are at doi:10.5281/zenodo.21199026. Seeds and hyperparameters as in §3.